

DroneArjuna (DA)

AI-Adjusted Project Timeline

Replanned task register assuming Claude-assisted development

ACS Technologies Limited

Document Date: 6 June 2026

Classification: RESTRICTED / CONFIDENTIAL

62 tasks across 6 phases | ~9–10 weeks AI-adjusted | ~4–5× speedup vs human baseline

Table of Contents

1. Calibration Basis

From session benchmarking, five tasks that would each be a half-day to full-day without AI (ManualControlPanel, CameraWindow, LiveOpsPanel, ConnectModal port scan, PlanWorkspace 3-panel layout) were each completed in 10–20 minutes — roughly a 10–15× speedup on pure code-generation work. The speedup is not uniform across task types.

Task Type	Human Baseline	With AI	Multiplier
CRUD endpoints + model + schema	1–2 days	1–2 h	10×
DB model + Alembic migration	½ day	20 min	8×
Frontend component (forms, lists, panels)	½–1 day	15–30 min	12×
pytest suite (unit + integration)	1–2 days	1–2 h	8×
Complex algorithm (geofencing, deconfliction)	3–5 days	1–2 days	3×
Third-party library integration (Cesium, Celery)	1–2 weeks	3–5 days	3×
ML pipeline (YOLOv8, workers)	2–3 weeks	5–7 days	3×
Infrastructure (Docker, CI, nginx)	3–5 days	1–2 days	3×
SITL environment + hardware validation	1–2 weeks	1–2 weeks	1× (fixed)
Architecture decisions + review	—	—	1× (fixed)

2. Revised Effort Scale

All task estimates in this document use the following AI-adjusted effort symbols:

Symbol	With AI	Human Baseline
XS	15–30 min	½ day
S	1–2 h	1 day
M	3–6 h	2–3 days
L	1–2 days	4–7 days
XL	3–5 days	1–2 weeks
⌚ (fixed)	Human-gated	No change

3. Phase 1 — Exit Gate

Phase total: 3–4 calendar days (human baseline: ~2 weeks)

Note: SITL environment provisioning (P1-03) is the bottleneck, not code.

#	Task	AI Effort	Note
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#	Task	AI Effort	Note
P1-01	GET /api/drone-control/ports — enumerate serial + UDP/TCP SITL ports	S (1h)	Straightforward
P1-02	POST /api/drone-control/autoconnect — probe ports, first success	S (1h)	Depends P1-01
P1-03	ArduPilot SITL smoke test — bring up simulator, verify full stack	⌘ (2–3 days)	Env setup, not code
P1-04	pytest auth suite — login, refresh, expiry, RBAC per role	S (1.5h)	—
P1-05	pytest drone-control suite — connect, command, WS frame	S (1.5h)	Needs P1-03
P1-06	Tag v1.0.0, baseline DA-STD/STR docs	XS (20 min)	Admin

4. Phase 2 — Drone Master + Mission Planner

Phase total: 5 calendar days (human baseline: 6–8 weeks)

4A. Payload Type CRUD

Subtotal: ~6 hours

#	Task	AI Effort
P2-01	Payload/PayloadType models + migration 003	XS (20 min)
P2-02	POST/GET/PUT/DELETE /api/master/payload-types	S (1h)
P2-03	POST/GET/PUT/DELETE /api/master/payloads (instances)	S (1h)
P2-04	Replace inventory stub GET /api/inventory/payloads	XS (30 min)
P2-05	Weight compliance check in MissionValidator	XS (30 min)
P2-06	PayloadManager.tsx — full CRUD UI	S (1.5h)
P2-07	Payload assignment in DroneInstanceManager.tsx	XS (30 min)

4B. Configuration Templates

Subtotal: ~2 hours

#	Task	AI Effort
P2-08	DroneConfigTemplate model + migration 004	XS (20 min)
P2-09	GET/POST/PUT/DELETE /api/master/config-templates	S (1h)
P2-10	Config template picker in ConnectModal.tsx	XS (30 min)

4C. Mission Planner Hardening

Subtotal: ~3 days

#	Task	AI Effort	Note
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#	Task	AI Effort	Note
P2-11	Geofence draw & edit tools in MapCanvas.tsx	L (1.5 days)	Leaflet Draw + custom handles
P2-12	Geofence validation visual feedback on map	S (1h)	—
P2-13	Survey-grid wizard UI	M (4h)	—
P2-14	Waypoint actions UI per waypoint (action + params)	M (3h)	—
P2-15	Mission status workflow buttons	XS (30 min)	Backend endpoint exists
P2-16	Pre-flight simulation preview animate	M (4h)	—
P2-17	pytest flight suite	S (1.5h)	—

5. Phase 3 — Advanced Flight + Visualisation

Phase total: 10–12 calendar days (human baseline: 8–10 weeks)

Note: Cesium.js is the dominant cost — a 600k-line library with its own paradigm.

#	Task	AI Effort	Note
P3-01	Cesium.js integration — scaffold Map3DCanvas.tsx	XL (3–4 days)	Large novel library; expect iteration
P3-02	3D terrain draping for mission routes	L (1.5 days)	Depends P3-01
P3-03	3D live drone tracking (glTF model + telemetry feed)	M (4h)	Depends P3-01
P3-04	Telemetry replay player (timeline scrubber)	M (4h)	—
P3-05	Geofence enforcement — breach detection + auto-RTL	L (1.5 days)	Algorithm + RabbitMQ event
P3-06	Multi-drone airspace deconfliction	L (2 days)	Geometry-heavy
P3-07	Vessel-relative home point — live update from feed	M (3h)	—
P3-08	HF position feed end-to-end validation	M (4h)	—
P3-09	pytest advanced flight suite	S (1.5h)	—

6. Phase 4 — Drone Inventory (Knowledge Base)

Phase total: 7–9 calendar days (human baseline: 6–8 weeks)

Note: Elasticsearch is the dominant cost. P4-01–P4-05 and P4-07 can be completed in a single focused day.

#	Task	AI Effort	Note
P4-01	InventoryWorkspace.tsx — new 6th workspace tab	S (1.5h)	—
P4-02	Drone catalogue browser — card grid + facet filters	M (3h)	—
P4-03	Drone spec detail page + performance chart	M (3h)	—

#	Task	AI Effort	Note
P4-04	Side-by-side comparison view	S (1.5h)	API already exists
P4-05	Payload catalogue browser	M (2h)	Depends P2-06
P4-06	Elasticsearch integration (replace ILIKE)	XL (3–4 days)	New infra + index mapping
P4-07	Instant search bar (debounced)	XS (30 min)	—
P4-08	Threat system / ORBAT model + knowledge-base page	L (2 days)	Novel domain model

7. Phase 5 — Drone Analyst (AI/CV Pipeline)

Phase total: 12–14 calendar days (human baseline: 10–12 weeks)

Note: ML model environment (CUDA/GPU, model weights, Celery worker config) is the irreducible complexity.

#	Task	AI Effort	Note
P5-01	Persist analysis jobs to DB — AnalysisJob model + migration 005	XS (30 min)	—
P5-02	MinIO upload endpoint (POST /api/analyst/upload)	S (1.5h)	—
P5-03	Celery worker infrastructure — broker config, worker/ directory	L (1.5 days)	New infra; Docker Compose changes
P5-04	YOLOv8 object detection Celery task	L (2 days)	ML model setup + GPU config
P5-05	Change detection worker (pixel-diff algorithm)	L (1.5 days)	Depends P5-03
P5-06	Video frame extraction Celery task	M (4h)	Depends P5-03
P5-07	Analyst workspace UI — job submission form	M (3h)	—
P5-08	Detection results viewer — bounding box overlay	L (1.5 days)	Complex canvas rendering
P5-09	Change detection map layer in Plan/Fly workspaces	M (3h)	—
P5-10	Telemetry analytics dashboard — rich charts	M (3h)	API already exists
P5-11	Automated telemetry report PDF export	L (2 days)	—

8. Phase 6 — Hardening, Security + UX Polish

Phase total: 5–6 calendar days (human baseline: 6–8 weeks)

#	Task	AI Effort	Note
P6-01	Full pytest coverage pass — ≥80% line coverage	L (1.5 days)	—
P6-02	JWT refresh rotation + Redis token blacklist	S (1.5h)	—
P6-03	Rate limiting on auth (slowapi)	XS (30 min)	—
P6-04	Input sanitisation + SQL injection audit	M (3h)	—

#	Task	AI Effort	Note
P6-05	TLS/HTTPS nginx reverse proxy	S (1.5h)	—
P6-06	WCAG 2.1 AA pass — focus rings, aria, contrast	M (4h)	Needs human visual review
P6-07	React error boundaries per workspace	XS (30 min)	—
P6-08	Service worker + offline indicator	M (3h)	—
P6-09	Locust load test — 10 drones × telemetry + 3 missions	M (3h)	—
P6-10	Docker image hardening (non-root, read-only fs)	S (1.5h)	—
P6-11	GitHub Actions CI pipeline	M (3h)	—

9. Revised Summary

Phase	Tasks	AI-Adjusted	Human Baseline	Speedup
Phase 1 — Exit Gate	6	3–4 days	~2 weeks	3–4× (SITL bottleneck)
Phase 2 — Drone Master + Mission	17	5 days	6–8 weeks	~10×
Phase 3 — Advanced Flight + Viz	9	10–12 days	8–10 weeks	~4× (Cesium bottleneck)
Phase 4 — Drone Inventory	8	7–9 days	6–8 weeks	~6×
Phase 5 — Drone Analyst AI	11	12–14 days	10–12 weeks	~5× (ML env bottleneck)
Phase 6 — Hardening + Polish	11	5–6 days	6–8 weeks	~8×
TOTAL	62	~9–10 weeks	~38–48 weeks	~4–5×

10. Week-by-Week Sprint Plan

Week 1 — Phase 1 Exit Gate

- Mon–Tue: P1-01, P1-02 (ports + autoconnect) + P1-04 (auth tests)
- Wed–Fri: P1-03 SITL environment setup ← provision simulator, unblock P1-05
- Fri: P1-05 (control tests), P1-06 (v1.0.0 tag)

Milestone: Phase 1 closed

Week 2 — Phase 2 Backend

- Mon: P2-01→P2-05 (payload model + all backend CRUD) + P2-08→P2-09 (config template backend) — same sitting
- Tue: P2-06→P2-07 (PayloadManager UI), P2-10 (config picker)
- Wed–Thu: P2-11 (geofence draw tools) ← largest Phase 2 task
- Fri: P2-12, P2-13, P2-14, P2-15 (validation UI, survey wizard, waypoint actions, status buttons)

Milestone: Phase 2 backend + data layer closed

Week 3 — Phase 2 Completion + Phase 4 UI

- Mon: P2-16 (simulation preview), P2-17 (flight tests) → Phase 2 fully closed
- Tue–Fri: P4-01→P4-05, P4-07 (inventory UI — full knowledge base browser, spec pages, compare, search bar)

Milestone: Phase 4 UI closed (Elasticsearch deferred)

Weeks 4–5 — Cesium + Analyst Foundation

- P3-01: Cesium scaffold (4 days — largest remaining task)
- Parallel: P5-01 (persist jobs), P5-02 (MinIO upload), P5-07 (analyst workspace UI), P5-10 (telemetry dashboard)

Weeks 5–6 — Phase 3 Completion

- P3-02, P3-03: 3D terrain draping + live drone tracking
- P3-04: Telemetry replay player
- P3-05, P3-06: Geofence enforcement + multi-drone deconfliction
- P3-07, P3-08: HF position feed + vessel-relative home
- P3-09: Advanced flight test suite

Milestone: Phase 3 closed

Weeks 6–7 — Phase 5 ML Pipeline

- P5-03: Celery worker infrastructure
- P5-04, P5-05: YOLOv8 detection + change detection workers
- P5-06: Video frame extraction
- P5-08, P5-09: Detection results viewer + map layer
- P5-11: PDF report export

Milestone: Phase 5 closed

Weeks 7–8 — Phase 4 Completion

- P4-06: Elasticsearch integration (3–4 days)
- P4-08: Threat system / ORBAT model + knowledge-base page

Milestone: Phase 4 fully closed

Weeks 8–9 — Phase 6 Hardening

- P6-01→P6-11: Full hardening sprint (pytest coverage, JWT hardening, rate limiting, WCAG, CI pipeline)

Milestone: Phase 6 closed

Weeks 9–10 — Buffer + Integration Testing

- Buffer + integration testing + DA-STD / DA-STR baseline update
- Final QA on real SITL with all phases active simultaneously

Milestone: PROJECT COMPLETE

11. What AI Does Not Accelerate

The following bottlenecks remain at their original human-baseline duration regardless of AI coding assistance. These are the true constraints on delivery velocity.

Bottleneck	Why It Is Fixed
SITL environment	Downloading ArduPilot, configuring Docker networking, forwarding MAVLink ports — environment work, not code
GPU/ML environment	YOLOv8 needs CUDA drivers, model weights downloaded, Docker GPU pass-through configured
Elasticsearch cluster	Another Docker service, index mapping design, data migration run

Bottleneck	Why It Is Fixed
WCAG visual review	Must be done by a human looking at the rendered UI, not code inspection
Security audit	Penetration testing, threat modelling, manual code review beyond linting
Architecture decisions	Where to put geofence enforcement boundary, event schema for RabbitMQ — judgement calls
Integration debugging	When the SITL drone doesn't behave as documented, or Celery task silently fails — requires iteration on running systems

AI compresses the coding phase from months to weeks, but every task with an “environment setup” component or a “test with real hardware” component remains at its original duration. The project is not 10× faster overall — it is closer to 4–5×, with the remaining constraint being infrastructure provisioning, external system integration, and the irreducible complexity of novel third-party libraries such as Cesium.